

vax_11_780_mmu_sop_v1.json

```
{
  "id": "SOP-VAX-002",
  "title": "VAX-11/780 Memory Management (MMU) & Translation Buffer Diagnostic",
  "system": "VAX-11/780",
  "component": "KA780 Memory Management Unit",
  "author": "SymbioStreams Semantic Bridge",
  "source": "DEC VAX Maintenance Handbook (EK-VAXV2-HB-002)",
  "description": "Diagnostic logic for isolating Memory Management (MMU) faults and Translation Buffer (TB) parity errors on the VAX-11/780.",
  "steps": [
    {
      "step": 1,
      "action": "Translation Buffer (TB) Error Identification",
      "details": "When a 'TB Parity Error' occurs, the CPU enters a machine check. Examine Internal Processor Register (IPR) 58 (TBADR).",
      "check": "Record the virtual address and the side of the buffer (Group 0 or Group 1) that failed."
    },
    {
      "step": 2,
      "action": "Address Translation Verification",
      "details": "Manually verify the Page Table Entry (PTE) for the failing virtual address. Check the 'Valid' (V) bit and 'Modify' (M) bit.",
      "indicators": {
        "V=0": "Translation failed due to invalid page. Possible OS software bug.",
        "Parity bit mismatch": "Hardware failure in the TB RAM or the PTE cache."
      }
    },
    {
      "step": 3,
      "action": "Run MMU Micro-diagnostics",
      "details": "Load the MMU Micro-diagnostics (EVKAC series) from the console floppy. Run the tests in 'Loop on Error' mode.",
      "logic": "The EVKAC routines specifically stress the MMU's translation logic and protection code checking."
    },
    {
      "step": 4,
      "action": "Module Level Isolation",
      "details": "The EVKAC diagnostic will provide a halt address. Cross-reference this with the MMU Module Utilization Chart.",
      "remedy": "Isolate to the failing board (typically M8220 or M8221 for MMU control) and replace."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "TB Parity Light ON",
      "next": "Read TBADR to find failing virtual address."
    },
    {
      "condition": "Intermittent MMU Fault".
    }
  ]
}
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    "next": "Verify CPU +5V and +15V voltage levels."
  },
  {
    "condition": "Solid TB Failure",
    "next": "Replace M8220 Memory Interconnect module."
  }
]
}
```